

Lecture 3 Areal Data Models

Shiwei Lan¹

¹School of Mathematical and Statistical Sciences
Arizona State University

STP598 Spatiotemporal Analysis
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Areal Data

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Spatial
Problems

Exploratory data
analysis (EDA)

Markov random
fields

Conditionally
autoregressive
(CAR) models

Simultaneous
autoregressive
(SAR) models

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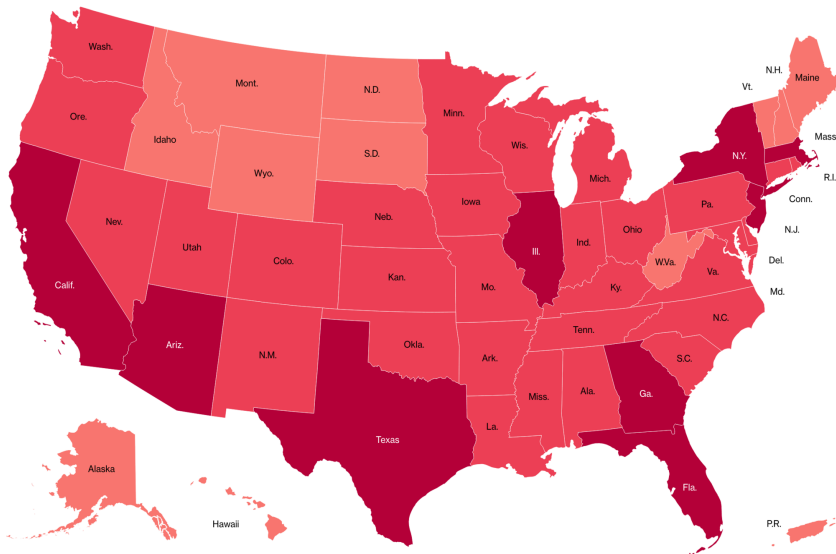
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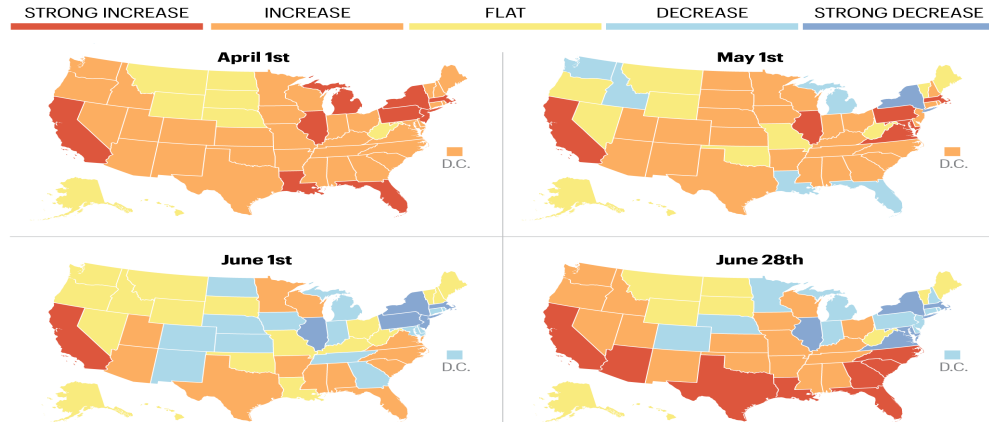
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RISING AND FALLING NEW CORONAVIRUS CASES

CHANGE IN DAILY NUMBER OF NEW CASES



SEVEN-DAY AVERAGE OF NEW CASES. "STRONG" CHANGE: IN EXCESS OF 500 CASES; "FLAT": +/- 25
 SOURCE: N.Y. TIMES COMPILATION OF STATE AND LOCAL GOVERNMENTS AND HEALTH DEPARTMENTS DATA

FORTUNE

In the context of areal units the general inferential issues are the following:

- ① Is there spatial pattern? If so, how strong is it?
- ② Do we want to smooth the data? If so, how much?
- ③ For a new areal unit or set of units, how can we infer about what data values we expect to be associated with these units? This is the so-called *modifiable areal unit problem (MAUP)*.

We will explore both descriptive and model-based approaches in this lecture.

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- The primary concept *proximity matrix* W for areal units $1, 2, \dots, n$ is defined by setting entries w_{ij} spatially connect units i and j ($w_{ii} = 0$).
- Binary choice: $w_{ij} = 1$ if i and j share common boundary; otherwise 0.
- 'Distance': e.g. decreasing function of intercentroidal distance between the units, binary values based on truncated distance or K nearest neighborhood.
- W can be standardized as $\widetilde{W}$ with $\widetilde{w}_{ij} = w_{ij}/w_{i+}$ where $w_{i+} = \sum_j w_{ij}$. $\widetilde{W}$ is row stochastic, i.e. $\widetilde{W}\mathbf{1} = \mathbf{1}$.
- Divide distances into bins $(0, d_1], (d_1, d_2], \dots$ and define k -th order neighbors of unit i as all units with distances in $(d_{k-1}, d_k]$. We can define k -th order proximity matrix $W^{(k)}$ based on k -th order neighbors.

Exploratory data analysis (EDA)

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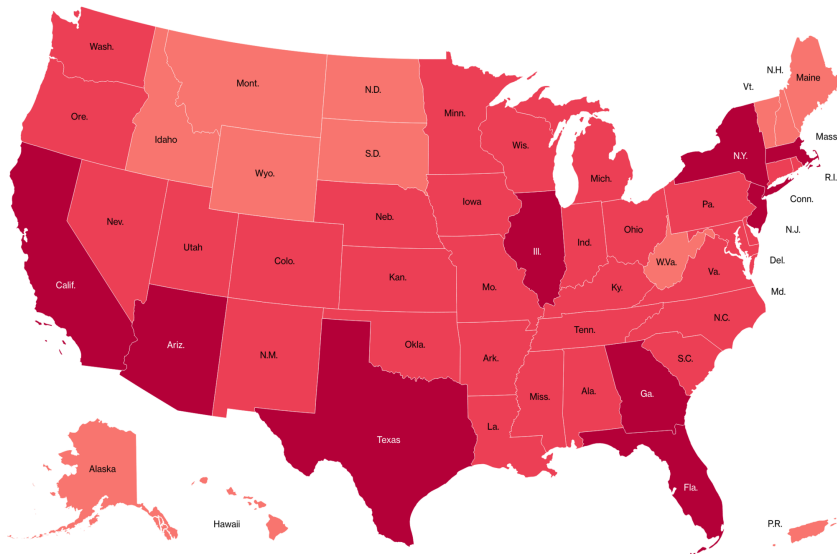
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- There are two standard statistics to measure the spatial association (Ripley, 1981).
- Moran's I :

$$I = \frac{n \sum_i \sum_j w_{ij} (Y_i - \bar{Y})(Y_j - \bar{Y})}{\left(\sum_{i \neq j} w_{ij} \right) \sum_i (Y_i - \bar{Y})^2} \quad (1)$$

- Under the null model where Y_i are i.i.d., $I \sim N(-1/(n-1), \text{Var}(I))$ with

$$\text{Var}(I) = \frac{n^2(n-1)S_1 - n(n-1)S_2 - 2S_0^2}{(n+1)(n-1)^2 S_0^2}$$

where $S_0 = \sum_{i \neq j} w_{ij}$, $S_1 = \frac{1}{2} \sum_{i \neq j} (w_{ij} + w_{ji})^2$, $S_2 = \sum_k (\sum_j w_{kj} + \sum_i w_{ik})^2$.

- Geary's C :

$$C = \frac{n \sum_i \sum_j w_{ij} (Y_i - Y_j)^2}{\left(\sum_{i \neq j} w_{ij} \right) \sum_i (Y_i - \bar{Y})^2} \quad (2)$$

- $C \sim N(1, \text{Var}(C))$ under the null model.

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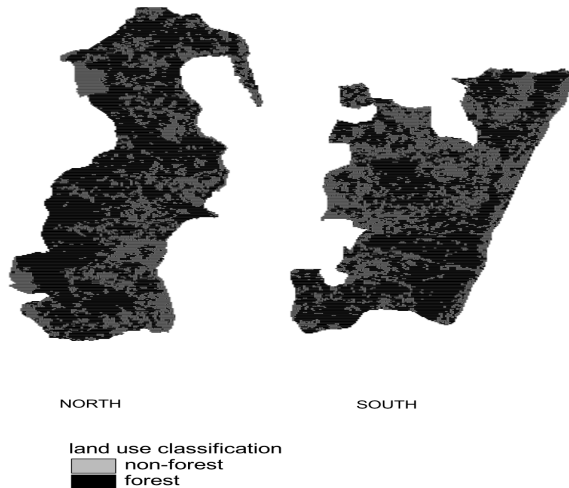


Figure 3.2 *Rasterized north and south regions ($1\text{ km} \times 1\text{ km}$) with binary land use classification overlaid.*

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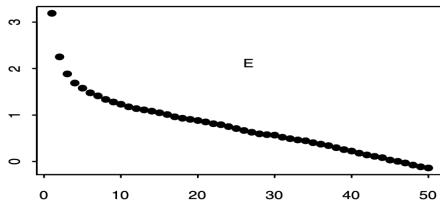
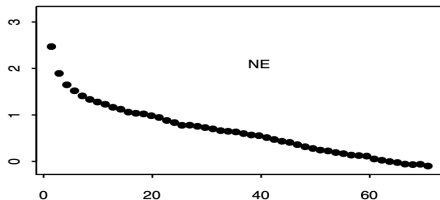
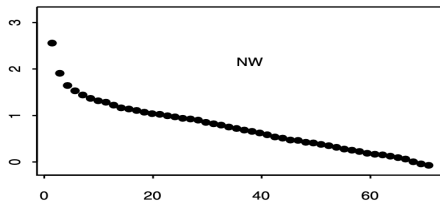
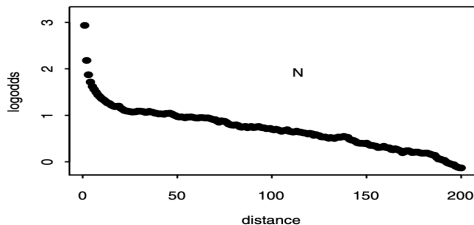


Figure 3.3 *Land use log-odds ratio versus distance in four directions.*

- One could also investigate a choropleth map by smoothing Y_i 's.
- The proximity matrix W provides a smoother: $\hat{Y}_i = \sum_j w_{ij} Y_j / w_{i+}$.
- However, $\hat{Y}_i$ ignores Y_i . We might revise it to be

$$\hat{Y}_i^* = (1 - \alpha) Y_i + \alpha \hat{Y}_i \quad (3)$$

where $\alpha \in (0, 1)$.

- One can refer to general *filters*.

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- Given $p(y_1, \dots, y_n)$, the so-called *full conditional* distributions, $p(y_i|y_j, j \neq i)$, $i = 1, \dots, n$, are uniquely determined.
- Brook's lemma (1964) proves the converse and constructively retrieve the unique joint distribution from these full conditionals.
- Compatible* conditionals, *proper* conditionals (improper joint).
- Brook's lemma:

$$p(y_1, \dots, y_n) = \frac{p(y_1|y_2, \dots, y_n)}{p(y_{10}|y_2, \dots, y_n)} \cdot \frac{p(y_2|y_{10}, y_3, \dots, y_n)}{p(y_{20}|y_{10}, y_3, \dots, y_n)} \dots \frac{p(y_n|y_{10}, \dots, y_{n-1,0})}{p(y_{n0}|y_{10}, \dots, y_{n-1,0})} \cdot p(y_{10}, \dots, y_{n0}) \quad (4)$$

- Denote ∂_i as the set of neighbors of unit i . An areal process Y_i is referred as a *Markov random field (MRF)* (Besag 1974, Kaiser and Cressie 2000) if

$$p(y_i|y_j, j \neq i) = p(y_i|y_j, j \in \partial_i) \quad (5)$$

- A *clique* is a set of cells (indices) such that each element is a neighbor of every other element.
- A *potential* of order k is a function of k exchangeable arguments.
- $p(y_1, \dots, y_n)$ is a *Gibbs distribution* if it is a function of the Y_i only through potentials on cliques:

$$p(y_1, \dots, y_n) \propto \exp \left\{ \gamma \sum_k \sum_{\alpha \in \mathcal{M}_k} \phi^{(k)}(y_{\alpha_1}, y_{\alpha_2}, \dots, y_{\alpha_k}) \right\} \quad (6)$$

where $\phi^{(k)}$ is a potential of order k , $\mathcal{M}_k$ is the collection of all subsets of size k from $\{1, 2, \dots, n\}$, $\alpha = (\alpha_1, \dots, \alpha_k)$ indexes this set, and $\gamma > 0$ is a scale parameter.

- The *Hammersley-Clifford Theorem* (Clifford 1990) demonstrates that if we have an MRF, then its joint distribution is a Gibbs distribution.
- Geman and Geman (1984) provides essentially the converse of the Hammersley-Clifford theorem: A Gibbs distribution determines an MRF.
- Sampling a MRF is reduced to sampling its associated Gibbs distribution, hence coining the term 'Gibbs sampler'.
- With cliques of order 1, we consider for continuous data on $\mathbb{R}^1$

$$p(y_1, \dots, y_n) \propto \exp \left\{ -\frac{1}{2\tau^2} \sum_{i,j} (y_i - y_j)^2 I(i \sim j) \right\} \quad (7)$$

- It is a Gibbs distribution on potentials of order 1 and 2 and that

$$p(y_i | y_j, j \neq i) = N \left(\sum_{j \in \partial_i} y_j / m_i, \tau^2 / m_i \right) \quad (8)$$

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- We begin with the Gaussian (*autonormal*) case. Suppose

$$Y_i | y_j, j \neq i \sim N \left(\sum_j b_{ij} y_j, \tau_i^2 \right), \quad i = 1, \dots, n \quad (9)$$

- By Brook's Lemma, we have

$$p(y_1, \dots, y_n) \propto \exp \left\{ -\frac{1}{2} \mathbf{y}' D^{-1} (I - B) \mathbf{y} \right\} \quad (10)$$

where $B = (b_{ij})$ and $D = \text{diag}\{\tau_i^2\}$.

- $\mathbf{Y} \sim N(\mathbf{0}, \Sigma_{\mathbf{y}} = (I - B)^{-1} D)$?

$$\frac{b_{ij}}{\tau_i^2} = \frac{b_{ji}}{\tau_j^2} \quad \text{for all } i, j \quad (11)$$

- Setting $b_{ij} = w_{ij}/w_{i+}$ and $\tau_i^2 = \tau^2/w_{i+}$, we have

$$p(y_i|y_j, j \neq i) = N \left(\sum_j w_{ij} y_j / w_{i+}, \tau^2 / w_{i+} \right) \quad (12)$$

- Therefore we have the joint distribution (intrinsically autoregressive, IAR)

$$p(y_1, \dots, y_n) \propto \exp \left\{ -\frac{1}{2\tau^2} \mathbf{y}'(D_w - W)\mathbf{y} \right\} = \exp \left\{ -\frac{1}{2\tau^2} \sum_{i \neq j} w_{ij} (y_i - y_j)^2 \right\} \quad (13)$$

where $D_w = \text{diag}\{w_{i+}\}$.

- $(D_w - W)\mathbf{1} = 0$. $\Sigma_y = ?$
- Redefine $\Sigma_y^{-1} = D_w - \rho W > 0$ for chosen $\rho \in (1/\lambda_{(1)}, 1/\lambda_{(n)})$.

- Rewriting the autonormal model

$$\mathbf{Y} = B\mathbf{Y} + \epsilon \quad (14)$$

- If $p(\mathbf{y})$ is proper, then:
 - $\mathbf{Y} \sim N(\mathbf{0}, (I - B)^{-1}D)$, $\epsilon \sim N(\mathbf{0}, D(I - B)^T)$, and $\text{Cov}(\epsilon, \mathbf{Y}) = D$.
 - $1/(\Sigma_{\mathbf{y}}^{-1})_{ii} = \text{Var}(Y_i | Y_j, j \neq i) = \tau_i^2$.
 - $(\Sigma_{\mathbf{y}}^{-1})_{ij} = b_{ij} = 0$ implies $Y_i \perp Y_j | Y_k, k \neq i, j$. We have control on conditional independence (by setting $w_{ij} = 0$)!
- One can introduce regression component to CAR.
- Considering a vector of dependent areal units leads to MCAR model.
- CAR model can be applied to point-level data.

- We could also consider non-Gaussian case.

$$p(y_i | y_j, j \neq i) = \exp(\{\psi(\theta_i y_i - \chi(\theta_i))\}) \quad (15)$$

where $\theta_i = \sum_{j \neq i} b_{ij} y_j$.

- Autologistic model:

$$\log \frac{P(Y_i = 1)}{P(Y_i = 0)} = \mathbf{x}_i^T \boldsymbol{\gamma} + \psi \sum w_{ij} y_j \quad (16)$$

which implies

$$p(y_1, \dots, y_n) \propto \exp \left(\boldsymbol{\gamma}^T \left(\sum_i y_i \mathbf{x}_i \right) + \psi \sum_{i,j} w_{ij} y_i y_j \right) \quad (17)$$

- Potts model

$$P(Y_i = l | Y_j, j \neq i) \propto \exp \left(\psi \sum_{i,j} w_{ij} I(Y_j = l) \right) \quad (18)$$

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- Now we start from $\epsilon \sim N(0, \tilde{D})$ with $\tilde{D} = \text{diag}\{\sigma_i^2\}$. Then

$$\mathbf{Y} = B\mathbf{Y} + \epsilon \sim N(\mathbf{0}, (I - B)^{-1}\tilde{D}(I - B)^{-T}) \quad (19)$$

where $\text{Cov}(\epsilon, \mathbf{Y}) = \tilde{D}(I - B)^{-1}$.

- $(I - B)$ must be full rank:
 - $B = \rho W$, W the contiguity matrix $w_{ij} = I(i \sim j)$. $Y_i = \rho \sum_j Y_j I(j \in \partial_i) + \epsilon_i$ with *spatial autoregressive parameter* $\rho \in (1/\lambda_{(1)}, 1/\lambda_{(n)})$.
 - $B = \alpha \tilde{W}$, with *spatial autocorrelation parameter* $\alpha \in (-1, 1)$.
- SAR model is introduced in a regression context and is applied to the *residuals* $\mathbf{U} = \mathbf{Y} - X\beta$:

$$\mathbf{U} = B\mathbf{U} + \epsilon \quad (20)$$

- The overall model is then written as follows with B interpolating between an OLS ($B = 0$) regression and a purely spatial model:

$$\mathbf{Y} = B\mathbf{Y} + (I - B)X\beta + \epsilon \quad (21)$$

- Assuming $\tilde{D} = \sigma^2 I$, the log-likelihood can be efficiently calculated thus amenable to MLE

$$\frac{1}{2} \log |\sigma^{-1}(I - B)| - \frac{1}{2\sigma^2} (\mathbf{Y} - X\beta)^T (I - B)(I - B)^T (\mathbf{Y} - X\beta) \quad (22)$$

- Extendable to Bayesian setting. No convenient form for full conditional distributions as in CAR.

- Both are spatial models for areal data.
- They are equivalent iff

$$(I - B)^{-1}D = (I - B)^{-1}\tilde{D}(I - B)^{-T} \quad (23)$$

- Cressie (1993) shows that any SAR model can be represented as a CAR model; but not vice versa.
- The first-order neighbor correlations increase at a slower rate as a function of ρ in the CAR model than in SAR model.
- Gibbs sampler is usually used for CAR but likelihood based inference is used for SAR.